

Verification phases

Phase 1	MATLAB reference		100%	50 frames, 8 algorithms, re-verified
Phase 2	HLS C-simulation		100%	4 designs, 50 frames, all bit-exact
Phase 3	Synthesis + RTL co-sim		100%	4 designs, exact latency, lossless in RTL
Phase 3b	Vivado place and route		100%	4 designs, replica of Kiem, 9 variants
Phase 4	On silicon		0%	needs board access

Thesis A objectives

Obj 1	Mitigate quantisation error		35%	root cause found; fix not yet implemented
Obj 2	Broaden algorithm comparison		65%	LPC and the no-prediction baseline done
Obj 3	Full compress + decompress chain		90%	verified both ways; routed; board remaining